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\usepackage{listings}
\usepackage{xcolor}

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\begin{document}

%=====================================================================
=========
% TITLE & METADATA
%=====================================================================
=========
\title{The Hilbert-Space Quadrature Conjecture:\\
Time as the Wave-Function-Generated Quadrature of Circular Phase}

\author{Steven Michael Allen Owens}
\affiliation{Independent Researcher}

\date{October 18, 2026}

%=====================================================================
=========
% ABSTRACT
%=====================================================================
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\begin{abstract}
We formulate and establish the conditional mathematical consistency of
the \emph{Hilbert-Space Quadrature Conjecture}, which proposes that
physical time is not a primitive external background parameter, but an
emergent, state-generated observable arising from the quadratic
mapping (``squaring'') of the intrinsic circular $U(1)$ phase structure
of Hilbert space. Let $H_U = H_C \otimes H_S$ be the bipartite Hilbert space of a
closed quantum universe containing a quantum clock $H_C$ and system $H_S$. To
avoid spectral commensurability singularities, exact relational
dynamics is formulated on the universal cover $H_C^{univ} = L^2(R, dt)$ via a
Gel'fand triple $\Phi \subset H_U \subset \Phi^\times$ and refined algebraic group averaging
$P_{phys} = \delta(\widehat{C})$. We prove that a continuous Bloch--Floquet winding quotient
map $\Pi_\omega : L^2(R) \to L^2(S^1)$ pushes the universal cover constraint forward to the
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compact clock circle S^1, where physical clocks with lower-bounded energy $\operatorname{Spec}(\widehat{N}) = \mathbb{N}_0$ reside in the Hardy space $H^2(S^1)$ and resolve the identity via a shift-covariant Holevo phase positive-operator-valued measure (POVM) $E(d\theta)$, thereby circumventing Pauli's theorem. We establish a tri-fold quadrature hierarchy distinguishing the raw clock marginal $M_\psi(\theta)$, the joint quadrature $J_\psi(\theta, F)$, and the conditional system state $\rho_S(\theta)$. In the ideal-clock limit, $\rho_S(\theta)$ satisfies the exact unitary Schr\"odinger equation, while approximate spectral-window wavepackets $|\Psi_w\rangle$ generate peaked semiclassical clock marginals. For realistic clocks with finite energy spread ΔE_C, the conditional quadrature convolution yields an exact Lindblad double-commutator master equation predicting intrinsic energy-basis dephasing at rate $\Gamma_{jk} \sim (E_j - E_k)^2/(2\hbar\Delta E_C)$. We prove the equivalence of Quadrature Time to Tomita--Takesaki modular flow, link the framework to Wigner--Weyl phase-space quantization, and extend the formalism to relativistic Klein--Gordon and Dirac constraints, stationary curved spacetimes (Schwarzschild gravitational time dilation, twin-shell decoherence, and horizon dephasing), minisuperspace cosmology, inflationary backreaction, Hartle--Hawking no-boundary states, and near-horizon Bisognano--Wichmann thermodynamics.
\end{abstract}

\maketitle

%==========
% SECTION I: INTRODUCTION
%==========
\section{Introduction: The Problem of External Time and the Phase Fiber}
\label{sec:intro}

In standard non-relativistic quantum mechanics, time $t \in R$ enters the Schr\"odinger equation as a classical background parameter external to the Hilbert space H. As proved by Pauli~\cite{Pauli1933}, any physical system whose Hamiltonian $\widehat{H}$ is semi-bounded below ($\operatorname{Spec}(\widehat{H}) \subset [E_0, \infty)$) admits no self-adjoint time operator $\widehat{T}$ satisfying the canonical Weyl commutation relation $[\widehat{T}, \widehat{H}] = i\hbar I$. Conversely, in canonical quantum gravity and reparameterization-invariant systems, the Wheeler--DeWitt constraint $\widehat{C}|\Psi\rangle = 0$ eliminates external time completely, leaving a globally stationary universal wave function

$|\Psi\rangle$~\cite{DeWitt1967}.

Two foundational programs have sought to recover temporal order from stationary quantum states:
\begin{enumerate}
    \item The \textbf{Page--Wootters mechanism}~\cite{PageWootters1983,Giovannetti2015}, which extracts relational dynamics from quantum entanglement between a clock subsystem H_C and an observed subsystem H_S.
    \item The \textbf{Connes--Rovelli thermal time hypothesis}~\cite{ConnesRovelli1994}, which identifies physical time flow with the state-dependent Tomita--Takesaki modular automorphism group σ_s^{Ψ} of a faithful density state on a von Neumann algebra of observables.
\end{enumerate}

In this paper, we unify and extend these approaches via a geometric and operational framework we call \textbf{State-Generated Quadrature Time} (or the \emph{Hilbert-Space Quadrature Conjecture}). Our starting point is the intrinsic geometry of complex Hilbert space. Normalized pure state vectors form the unit sphere $S(\mathcal{H}) = \{\psi \in \mathcal{H} : \|\psi\| = 1\}$, while physical rays reside in the projective Hilbert space $P(H) = S(H)/U(1)$ ~\cite{Kibble1979,Ashtekar1999}. This quotient defines a principal circular fiber bundle:
\begin{equation}
    U(1) \hookrightarrow S(\mathcal{H}) \xrightarrow{\pi} \mathbb{P}(\mathcal{H}).
    \label{eq:hopf_bundle}
\end{equation}
Along a single ray, the global circular phase $\psi \sim e^{i\theta}\psi$ ($\theta \in S^1 \cong U(1)$) is gauge-redundant. However, when a universal state $|\Psi\rangle \in \mathcal{H}_C \otimes \mathcal{H}_S$ entangles a clock sector with a system sector, the quadratic Born map---and its phase-space extension, the Wigner--Weyl transform~\cite{Wigner1932}---``squares'' the circular phase amplitudes of $|\Psi\rangle$ into a nonnegative real measure over clock readings and conditional system states. In this picture, \emph{the wave function does not merely evolve within time; the wave function of the clock sector is the physical generator that squares the circular phase geometry of Hilbert space into observable temporal order.}

To render this conjecture mathematically rigorous, our architecture resolves four foundational tensions:
\begin{enumerate}

\item \textbf{Spectral Singularity Resolution (Sec.~\ref{sec:universal_cover}):} We formulate the exact relational constraint $\widehat{C}|\Psi\rangle = 0$ on the universal cover R via a Gel'fand triple $\Phi \subset H_U \subset \Phi^\times$ and group averaging, and prove that a Bloch--Floquet winding quotient map Π_ω connects R equivariantly to the compact clock circle S^1.

\item \textbf{Pauli's Theorem Avoidance (Sec.~\ref{sec:povm}):} By restricting the physical clock to the lower-bounded Hardy space $H^2(S^1)$ ($\operatorname{Spec}(\widehat{N}) = \mathbb{N}_0$), clock readings are represented by a shift-covariant Holevo phase POVM $E(d\theta)$ on S^1 rather than a sharp self-adjoint operator~\cite{Holevo1982,SaleckerWigner1958,PeggBarnett1989}.

\item \textbf{Tri-Fold Quadrature Separation (Sec.~\ref{sec:quadrature}):} We distinguish the raw clock marginal $M_\Psi(\theta)$ (which is uniform for exact constraint eigenstates and peaked for spectral-window wavepackets $|\Psi_w\rangle$), the joint quadrature $J_\Psi(\theta, F)$, and the conditional system state $\rho_S(\theta)$.

\item \textbf{Finite-Clock Lindblad Master Equation (Sec.~\ref{sec:lindblad}):} We prove that finite clock energy spread ΔE_C induces a Gaussian phase convolution whose Markovian limit is a completely positive double-commutator master equation with a testable dephasing rate $\Gamma_{jk} \sim (E_j - E_k)^2/(2\hbar\Delta E_C)$.

\end{enumerate}

%===
% SECTION II: UNIVERSAL COVER, RIGGED HILBERT SPACE & WINDING QUOTIENT
%===
\section{Exact Relational Dynamics on the Universal Cover and the Winding Quotient Map}
\label{sec:universal_cover}

\subsection{The Spectral Obstruction on Compact S^1}
Suppose one defines the clock Hilbert space directly on the circle as $H_C = L^2(S^1, d\theta/2\pi)$ with discrete integer spectrum $\widehat{N}|n\rangle = n|n\rangle$ ($n \in Z$) and imposes the exact constraint

\begin{equation}
\widehat{C}_{\mathrm{naive}}|\Psi\rangle = \left(\hbar\omega \widehat{N} \otimes I_S + I_C \otimes \widehat{H}_S\right)|\Psi\rangle = 0.
\label{eq:naive_constraint}
\end{equation}

In the joint eigenbasis $\{|n\rangle_C \otimes |E_k\rangle_S\}$, Eq.~\eqref{eq:naive_constraint} requires $\hbar\omega n + E_k = 0$, which admits nonzero solutions in $H_C \otimes H_S$ if and only if every system energy eigenvalue satisfies the strict commensurability condition $E_k/(\hbar\omega) \in Z$. Equivalently, strict periodicity $|\Psi(2\pi)\rangle = |\Psi(0)\rangle$ forces $e^{-i2\pi\widehat{H}_S/(\hbar\omega)} = I_S$ on the physical subspace. For a generic Hamiltonian $\widehat{H}_S$ with incommensurate energy gaps or continuous spectrum, $\ker(\widehat{C}_{naive}) = \{0\}$ in the strict Hilbert space.

\subsection{Resolution A: Exact Dynamics on the Universal Cover via Group Averaging}
To eliminate this spectral obstruction, we formulate the exact kinematical constraint on the universal cover of the circle,
\begin{equation}
    \mathcal{H}_C^{\mathrm{univ}} = L^2(\mathbb{R}, dt),
\end{equation}
equipped with generalized time eigenstates $|t\rangle$ satisfying $\langle t | t' \rangle = \delta(t - t')$ and self-adjoint clock momentum operator $\widehat{P}_t = -i\hbar\frac{d}{dt}$ with continuous spectrum $\operatorname{Spec}(\widehat{P}_t) = \mathbb{R}$. The universal constraint operator on $H_U^{kin} = H_C^{univ} \otimes H_S$ is
\begin{equation}
    \widehat{C}_{\mathrm{univ}} = \widehat{P}_t \otimes I_S + I_C \otimes \widehat{H}_S.
    \label{eq:univ_constraint}
\end{equation}
Because zero lies in the continuous spectrum of $\widehat{C}_{univ}$, physical states are distributional elements of a Gel'fand triple (rigged Hilbert space)~\cite{Marolf1995}:
\begin{equation}
    \Phi \subset \mathcal{H}_U^{\mathrm{kin}} \subset \Phi^\times,
\end{equation}
where $\Phi = S(R) \otimes D(\widehat{H}_S)$ is the dense nuclear space of rapidly decreasing smooth test states. Physical states $|\Psi\rangle \in \Phi^\times$ are constructed via refined algebraic quantization (group averaging) through the rigging map $P_{phys}: \Phi \to \Phi^\times$:
\begin{equation}
    \mathbb{P}_{\mathrm{phys}} = \delta(\widehat{C}_{\mathrm{univ}}) = \frac{1}{2\pi\hbar} \int_{-\infty}^{\infty} ds \, e^{i s \widehat{C}_{\mathrm{univ}} / \hbar},
    \label{eq:group_averaging}

\end{equation}
which induces the positive-definite physical inner product on the constraint surface:
\begin{equation}
    \langle \phi | \psi \rangle_{\mathrm{phys}} := \int_{-\infty}^{\infty} \frac{ds}{2\pi\hbar} \, \langle \phi | e^{i s \widehat{C}_{\mathrm{univ}} / \hbar} | \psi \rangle_{\mathrm{kin}}.
    \label{eq:phys_inner_product}
\end{equation}
In the generalized clock basis $|t\rangle$, any group-averaged state $|\Psi\rangle = \mathbb{P}_{\mathrm{phys}}|\psi\rangle$ satisfies $\langle t | \widehat{C}_{\mathrm{univ}} | \Psi \rangle = 0$, or
\begin{equation}
    \left(-i\hbar \frac{\partial}{\partial t} + \widehat{H}_S\right) |\Psi(t)\rangle = 0 \implies |\Psi(t)\rangle = e^{-i \widehat{H}_S t / \hbar} |\Psi(0)\rangle,
    \label{eq:univ_evolution}
\end{equation}
for arbitrary system spectra $\operatorname{Spec}(\widehat{H}_S) \subset \mathbb{R}$.

\subsection{The Bloch--Floquet Winding Quotient Map Connecting R to S^1}
To connect the universal cover $H_C^{univ} = L^2(R, dt)$ to an operational cyclic clock on the circle $S^1 \cong \mathbb{R} / (T_0 \mathbb{Z})$ with fundamental period $T_0 = 2\pi/\omega$, we introduce the Bloch--Floquet winding quotient map.

\begin{proposition}[Bloch--Floquet Winding Quotient Homomorphism]
\label{prop:winding}
Let $H_C^{univ} = L^2(R, dt)$ be the universal cover clock Hilbert space with momentum operator $\widehat{P}_t = -i\hbar\frac{d}{dt}$ and generalized basis $\{|t\rangle\}_{t \in \mathbb{R}}$, and let $H_C^{circ} = H^2(S^1) \subset L^2(S^1, d\theta/2\pi)$ be the Hardy space on the circle of period $T_0 = 2\pi/\omega$ with number generator $\widehat{N} = -i\frac{\partial}{\partial\theta}$ ($\operatorname{Spec}(\widehat{N}) = \mathbb{N}_0$) and clock Hamiltonian $\widehat{H}_C = \hbar\omega\widehat{N}$. Then:
\begin{enumerate}
    \item The periodization map $\Pi_\omega : S(R) \to C^\infty(S^1)$ defined over winding sectors $m \in Z$ by
    \begin{equation}
        (\Pi_\omega f)(\theta) := \sum_{m=-\infty}^{\infty} f\!\left(\frac{\theta + 2\pi m}{\omega}\right), \qquad \theta \in [0, 2\pi),

\label{eq:winding_map}
\end{equation}
intertwines temporal translations on R with $U(1)$ rotations on S^1.
\item Under the sign convention $\theta = -\omega t \pmod{2\pi}$ (or conjugated clock phase), $\Pi_\omega \otimes I_S$ maps the universal cover constraint $\widehat{C}_{univ} = \widehat{P}_t \otimes I_S + I_C \otimes \widehat{H}_S$ equivariantly onto the circular constraint operator
\begin{equation}
\widehat{C}_{\mathrm{circ}} = \hbar\omega \widehat{N} \otimes I_S - I_C \otimes \widehat{H}_S.
\label{eq:circ_constraint}
\end{equation}
\item Summing the universal cover generalized states $|t\rangle$ across all winding sectors $m \in Z$ projected onto the positive-frequency Hardy space $H^2(S^1)$ yields the covariant Holevo phase POVM $E(d\theta) = \frac{1}{2\pi}|\theta)(\theta|d\theta$ on S^1.
\end{enumerate}
\end{proposition}

\begin{proof}
For any Schwartz test function $f \in S(R)$, rapid decay ensures absolute and uniform convergence of the series in Eq.~\eqref{eq:winding_map} on $[0, 2\pi)$, defining a smooth 2π-periodic function on S^1. A shift by $\tau \in R$, $(T_\tau f)(t) = f(t - \tau)$, maps under Π_ω to
\begin{equation}
(\Pi_\omega T_\tau f)(\theta) = \sum_{m=-\infty}^{\infty} f\!\left(\frac{\theta - \omega\tau + 2\pi m}{\omega}\right) = (\Pi_\omega f)(\theta - \omega\tau \pmod{2\pi}),
\end{equation}
proving that Π_ω intertwines the R-action with the $U(1)$-action on S^1. Conversely, given a smooth partition-of-unity window function $w(t) \in C_c^\infty(R)$ supported within a single fundamental domain $(-T_0/2, T_0/2)$ satisfying $\sum_{m\in Z} |w(t - mT_0)|^2 = 1$, the local section lift $(\Pi_\omega^* g)(t) = g(\omega t \pmod{2\pi})\, w(t)$ embeds $L^2(S^1)$ isometrically into a single winding sector of $L^2(R)$.

To establish the push-forward of the constraint operator, differentiate Eq.~\eqref{eq:winding_map} with respect to $\theta = \omega t$ using the chain rule $\frac{\partial}{\partial\theta} = \frac{1}{\omega}\frac{\partial}{\partial t}$. For a universal cover solution satisfying $-i\hbar \frac{\partial}{\partial t}|\Psi_{\mathrm{univ}}(t)\rangle + \widehat{H}_S\, |\Psi_{\mathrm{univ}}(t)\rangle = 0$, projecting onto

the positive-frequency circular sector where $|\Psi_{\mathrm{circ}}(\theta)\rangle = \sum_{n=0}^\infty e^{-in\theta}|\Psi_n\rangle$ (so that the conjugate phase bra $\langle\theta| = \sum_{n=0}^\infty e^{-in\theta}\langle n|$ extracts positive clock modes $n \geq 0$) yields

\begin{equation}
    i\hbar\omega \frac{\partial}{\partial \theta} |\Psi_{\mathrm{circ}}(\theta)\rangle = \widehat{H}_S |\Psi_{\mathrm{circ}}(\theta)\rangle \iff \left(\hbar\omega \widehat{N} \otimes I_S - I_C \otimes \widehat{H}_S\right)|\Psi_{\mathrm{circ}}\rangle = 0.
\end{equation}

Finally, partitioning $t \in R$ into its circular phase $\theta = \omega t \pmod{2\pi}$ and integer winding number $m = \lfloor \omega t / 2\pi \rfloor$, the sum of generalized eigenstates over all winding sectors defines the periodic Dirac comb distribution

\begin{equation}
    |\theta_{\mathrm{comb}}\rangle = \sum_{m=-\infty}^{\infty} \left| t = \frac{\theta + 2\pi m}{\omega} \right\rangle = \sum_{n=-\infty}^{\infty} e^{in\theta} |n\rangle.
\end{equation}

Applying the orthogonal Hardy projection $\mathbb{P}_+ = \sum_{n=0}^\infty |n\rangle\langle n|$ onto physical lower-bounded clock states ($n \geq 0$) maps $|\theta_{\mathrm{comb}}\rangle \mapsto |\theta) = \sum_{n=0}^\infty e^{in\theta}|n\rangle$, wrapping the Lebesgue measure dt into the covariant phase POVM $E(d\theta) = \frac{1}{2\pi}|\theta)(\theta|d\theta$.

\end{proof}

\subsection{Resolution B: Operational Compactifications on S^1}

When working directly on the compact clock space H_C^{circ} without passing to the full universal cover, spectral incommensurability is resolved operationally in one of two ways:

\begin{enumerate}
    \item \textbf{Spectral Window Constraint:} Replace the sharp equality $\widehat{C}|\Psi\rangle = 0$ with a bounded variance condition $\langle \Psi | \widehat{C}^2 | \Psi \rangle \le \epsilon^2$ using a regularized window filter $f(\widehat{C}/\epsilon)$ peaked at zero, yielding normalizable states $|\Psi_\epsilon\rangle = f(\widehat{C}/\epsilon)|\eta\rangle / \|f(\widehat{C}/\epsilon)|\eta\rangle\|$.
    \item \textbf{Finite Pegg--Barnett Clock ($H_C^{(N)} = C^N$):} Truncate the clock Hilbert space to N energy levels $\{|0\rangle, \dots, |N-1\rangle\}$ with discrete orthonormal phase states~\cite{PeggBarnett1989}
    \begin{equation}

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        |\theta_k^{(N)}\rangle = \frac{1}{\sqrt{N}} \sum_{n=0}^{N-1}
e^{in\theta_k} |n\rangle, \qquad \theta_k = \frac{2\pi k}{N}, \quad k
\in \{0, 1, \dots, N-1\}.
        \label{eq:pegg_barnett_states}
    \end{equation}
\end{enumerate}

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% SECTION III: COVARIANT CLOCK PHASE POVM
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\section{The Covariant Clock Phase POVM and Avoidance of Pauli's
Theorem}
\label{sec:povm}

Let the physical clock Hilbert space be the Hardy space $H_C = H^2(S^1)$
spanned by orthonormal energy eigenstates $\{|n\rangle\}_{n=0}^\infty$
with lower-bounded Hamiltonian $\widehat{H}_C = \hbar\omega\widehat{N}$
($\operatorname{Spec}(\widehat{N}) = \mathbb{N}_0$). Following
Holevo~\cite{Holevo1982}, we define the generalized circular phase
vectors $|\theta) \in \Phi_C^\times$ for $\theta \in [0, 2\pi) \cong S^1$:
\begin{equation}
    |\theta) = \sum_{n=0}^{\infty} e^{in\theta} |n\rangle.
    \label{eq:holevo_vectors}
\end{equation}
Because the sum is bounded below by $n = 0$, the states $|\theta)$ are
non-orthogonal. Their distributional inner product is the boundary
Szeg\H{o} kernel on the unit disk:
\begin{equation}
    (\theta | \theta') = \sum_{n=0}^{\infty} e^{in(\theta' - \theta)}
= \pi \delta_{2\pi}(\theta - \theta') + \frac{1}{1 - e^{i(\theta' -
\theta + i0^+)}}.
    \label{eq:szego_kernel}
\end{equation}

\begin{proposition}[Covariant Circular Clock POVM]
\label{prop:povm}
The operator-valued measure $E: B(S^1) \to B(H_C)$ defined on Borel subsets
$\Delta \subset S^1$ by
\begin{equation}
    E(\Delta) = \int_\Delta E(d\theta), \qquad E(d\theta) :=
\frac{1}{2\pi} |\theta)(\theta| \, d\theta,
    \label{eq:clock_povm}
\end{equation}
```

is a normalized, shift-covariant positive-operator-valued measure (POVM) satisfying:
\begin{enumerate}
    \item \textbf{Positivity:} $\langle \chi | E(\Delta) | \chi \rangle = \int_\Delta \frac{d\theta}{2\pi} |(\theta|\chi\rangle|^2 \ge 0$ for all $|\chi\rangle \in \mathcal{H}_C$.
    \item \textbf{Completeness:} $E(S^1) = \int_0^{2\pi} \frac{d\theta}{2\pi} |\theta)(\theta| = I_C$.
    \item \textbf{Phase Shift Covariance:} For the unitary clock generator $U(\phi) = e^{i\phi\widehat{N}}$,
    \begin{equation}
        U(\phi) E(\Delta) U^\dagger(\phi) = E(\Delta \oplus \phi), \qquad \forall \phi \in S^1,
        \label{eq:povm_covariance}
    \end{equation}
    where $\oplus$ denotes addition modulo 2π.
    \item \textbf{Finite-N Weak Convergence:} The finite-dimensional Pegg--Barnett projection-valued measure $E^{(N)}(\Delta) = \sum_{\theta_k \in \Delta} |\theta_k^{(N)}\rangle\langle\theta_k^{(N)}|$ converges in the weak operator topology to $E(\Delta)$ as $N \to \infty$.
\end{enumerate}
\end{proposition}

\begin{proof}
Positivity is immediate from $|(\theta|\chi\rangle|^2 \ge 0$. For completeness, let $|\chi\rangle = \sum_{n=0}^\infty c_n |n\rangle \in \mathcal{H}_C$. By Parseval's theorem for Fourier series in $H^2(S^1)$,
\begin{equation}
    \langle \chi | E(S^1) | \chi \rangle = \int_0^{2\pi} \frac{d\theta}{2\pi} \sum_{n=0}^{\infty} \sum_{m=0}^{\infty} c_n^* c_m e^{i(n-m)\theta} = \sum_{n=0}^{\infty} \sum_{m=0}^{\infty} c_n^* c_m \delta_{nm} = \sum_{n=0}^{\infty} |c_n|^2 = \langle \chi | \chi \rangle,
\end{equation}
so $E(S^1) = I_C$ by polarization. For phase shift covariance, observe that $U(\phi)|\theta) = \sum_{n=0}^\infty e^{in\phi} e^{in\theta}|n\rangle = |\theta + \phi)$, whence changing variables $\theta' = \theta + \phi \pmod{2\pi}$ in Eq.~\eqref{eq:clock_povm} yields $U(\phi)E(\Delta)U^\dagger(\phi) = E(\Delta \oplus \phi)$. Finally, for any $|\chi\rangle, |\eta\rangle \in \mathcal{H}_C$ and Borel interval $\Delta \subset S^1$, the Riemann sum
\begin{equation}
    \langle \chi | E^{(N)}(\Delta) | \eta \rangle = \frac{1}{N} \sum_{\theta_k \in \Delta} \left(\sum_{n=0}^{N-1} c_n^* e^{-in\theta_k}\right)\left(\sum_{m=0}^{N-1} d_m e^{im\theta_k}\right)

```
\xrightarrow{N \to \infty} \int_\Delta \frac{d\theta}{2\pi} \langle
\chi | \theta)(\theta | \eta \rangle = \langle \chi | E(\Delta) | \eta
\rangle,
\end{equation}
establishing weak operator convergence.
\end{proof}

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% SECTION IV: QUADRATURE CONJECTURE & CONDITIONAL DYNAMICS
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\section{The Quadrature Conjecture and Tri-Fold Conditional Structure}
\label{sec:quadrature}

To avoid conflating global stationarity with conditional evolution, we
explicitly separate three distinct quadratic objects generated by the
universal state $|\Psi\rangle$.

\begin{definition}[Tri-Fold Quadrature Architecture]
\label{def:trifold}
Let $|\Psi\rangle \in \mathcal{H}_C \otimes \mathcal{H}_S$ be a state
of the composite universe, let $E(d\theta) = \frac{1}{2\pi}|\theta)(\theta|d\theta$ be the covariant clock
POVM on $S^1$, and let $|\Psi(\theta)\rangle := ((\theta|_C \otimes
I_S)|\Psi\rangle \in \mathcal{H}_S$ be the unnormalized conditional
system vector. We define:
\begin{enumerate}
    \item \textbf{The Raw Clock Marginal ($M_\Psi$):} The probability
density for the clock to register phase $\theta \in S^1$ regardless of the system
state:
    \begin{equation}
        M_\Psi(\theta) := \frac{1}{2\pi}
\|\Psi(\theta)\|_{\mathcal{H}_S}^2 = \frac{1}{2\pi} \langle \Psi |
\left(|\theta)(\theta|_C \otimes I_S\right) | \Psi \rangle, \qquad
T_\Psi(d\theta) = M_\Psi(\theta)\,d\theta.
        \label{eq:raw_marginal}
    \end{equation}
    \item \textbf{The Joint Clock-System Quadrature ($J_\Psi$):} For any
system effect or projector $F \in B(H_S)$ $(0 \le F \le I_S)$, the joint probability
density of observing system property $F$ at clock reading $\theta$:
    \begin{equation}
        J_\Psi(\theta, F) := \frac{1}{2\pi} \langle \Psi(\theta) | F |
\Psi(\theta) \rangle = \frac{1}{2\pi} \langle \Psi |
\left(|\theta)(\theta|_C \otimes F\right) | \Psi \rangle.
        \label{eq:joint_quadrature}
    \end{equation}
```

\item \textbf{The Conditional System State ($\rho_S(\theta)$):} The normalized density operator of the system conditioned on clock phase θ:
\begin{equation}
\rho_S(\theta) := \frac{\operatorname{Tr}_C\left[(E(d\theta) \otimes I_S)|\Psi\rangle\langle\Psi|\right]}{\langle \Psi | E(d\theta) \otimes I_S | \Psi \rangle} = \frac{|\Psi(\theta)\rangle\langle\Psi(\theta)|}{\||\Psi(\theta)\|_{\mathcal{H}_S}^2},
\label{eq:cond_state}
\end{equation}
such that the conditional probability of outcome F given clock reading θ is $P_\Psi(F | \theta) = J_\Psi(\theta, F)/M_\Psi(\theta) = \operatorname{Tr}_S[\rho_S(\theta) F]$.
\end{enumerate}
\end{definition}

\begin{conjecture}[The Hilbert-Space Quadrature Conjecture]
\label{conj:main}
In a closed quantum universe $H_U = H_C \otimes H_S$, there exists no primitive external background time parameter. All observable temporal ordering and dynamical evolution are generated internally by the universal wave function $|\Psi\rangle$ through the quadratic mapping $\Psi \mapsto (M_\Psi, J_\Psi, \rho_S)$ over the covariant circular phase POVM $E(d\theta)$. Classical time $t = \theta/\omega$ is a derived semiclassical statistic of this state-generated quadrature structure.
\end{conjecture}

\begin{theorem}[Emergence of Conditional Schr\"odinger Dynamics]
\label{thm:schrodinger}
Let $|\Psi\rangle \in \Phi^\times$ be an exact physical state satisfying the relational constraint $\widehat{C}_{\mathrm{circ}}|\Psi\rangle = \left(\hbar\omega \widehat{N} \otimes I_S - I_C \otimes \widehat{H}_S\right)|\Psi\rangle = 0$ on the positive-spectrum subspace where $E_k = \hbar\omega n_k$ $(n_k \in N_0)$, or equivalently let $|\Psi(t)\rangle$ be the universal cover state of Eq.~\eqref{eq:univ_evolution} within a single fundamental period $t = \theta/\omega \in [0, 2\pi/\omega)$. Then:
\begin{enumerate}
\item The raw clock marginal is globally uniform across the circle:
\begin{equation}
M_\Psi(\theta) = \frac{1}{2\pi} = \mathrm{const.}
\label{eq:uniform_marginal}
\end{equation}
\item The conditional system state $\rho_S(t) := \rho_S(\omega t)$ is pure, $\rho_S(t) = |\psi(t)\rangle\langle\psi(t)|$, and satisfies the exact unitary

Schr\"odinger--von Neumann equation:
    \begin{equation}
        i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle =
\widehat{H}_S |\psi(t)\rangle, \qquad i\hbar \frac{d}{dt} \rho_S(t) =
[\widehat{H}_S, \rho_S(t)].
        \label{eq:schrodinger_emergent}
    \end{equation}
\end{enumerate}
\end{theorem}

\begin{proof}
Expanding $|\Psi\rangle = \sum_{n=0}^\infty \sum_k c_{n,k} |n\rangle_C
\otimes |E_k\rangle_S$ and imposing
$\widehat{C}_{\mathrm{circ}}|\Psi\rangle = 0$ yields $(\hbar\omega n - E_k)c_{n,k} = 0$,
restricting the support to $n = n_k = E_k/(\hbar\omega)$ with normalized coefficients
$\sum_k |\alpha_k|^2 = 1$:
\begin{equation}
    |\Psi\rangle = \sum_k \alpha_k |n_k\rangle_C \otimes
|E_k\rangle_S.
    \label{eq:entangled_stationary}
\end{equation}
Projecting onto the conjugate phase vector $(\theta|_C =
\sum_{m=0}^\infty e^{-im\theta}\langle m|_C$ gives the conditional
amplitude
\begin{equation}
    |\Psi(\theta)\rangle = ((\theta|_C \otimes I_S)|\Psi\rangle =
\sum_k \alpha_k e^{-i n_k \theta} |E_k\rangle_S = \sum_k \alpha_k
e^{-i E_k \theta / (\hbar\omega)} |E_k\rangle_S.
    \label{eq:psi_theta_expansion}
\end{equation}
Taking the norm squared of Eq.~\eqref{eq:psi_theta_expansion} using
the orthonormality $\langle E_j | E_k \rangle_S = \delta_{jk}$ yields
\begin{equation}
    M_\Psi(\theta) = \frac{1}{2\pi} \langle \Psi(\theta) |
\Psi(\theta) \rangle = \frac{1}{2\pi} \sum_k |\alpha_k|^2 =
\frac{1}{2\pi},
\end{equation}
which proves global stationarity of the raw marginal. Setting $t = \theta/\omega$ ($\frac{\partial}{\partial\theta} = \frac{1}{\omega}\frac{\partial}{\partial t}$) and differentiating $|\psi(t)\rangle = |\Psi(\omega
t)\rangle$ gives
\begin{equation}
    i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle = i\hbar\omega
\frac{\partial}{\partial \theta} \sum_k \alpha_k e^{-i E_k \theta /
(\hbar\omega)} |E_k\rangle_S = \sum_k \alpha_k E_k e^{-i E_k t /
\hbar} |E_k\rangle_S = \widehat{H}_S |\psi(t)\rangle,

\end{equation}
from which $i\hbar\frac{d}{dt}\rho_S(t) = [\widehat{H}_S, \rho_S(t)]$ follows immediately.
\end{proof}

\begin{proposition}[Semiclassical Peaked Time via Spectral-Window Wavepackets]
\label{prop:wavepacket}

Let $w(t) \in S(R)$ be a normalized clock wavepacket ($\int_R |w(t)|^2 dt = 1$) peaked around a semiclassical time t_0 with temporal width ΔT, and define the normalizable history state on $H_C^{univ} \otimes H_S$:
\begin{equation}
    |\Psi_w\rangle = \int_{-\infty}^{\infty} dt \, w(t) \, |t\rangle_C \otimes e^{-i\widehat{H}_S t / \hbar} |\psi_0\rangle_S.
    \label{eq:wavepacket_state}
\end{equation}
Then $|\Psi_w\rangle$ satisfies the spectral-window constraint bound
\begin{equation}
    \|\widehat{C}_{\mathrm{univ}}} |\Psi_w\rangle\|_{\mathcal{H}_U} = \hbar \|w'\|_{L^2(\mathbb{R})} \sim \frac{\hbar}{\Delta T},
    \label{eq:window_bound}
\end{equation}
its raw clock marginal is the peaked distribution $M_{\Psi_w}(t) = |w(t)|^2$, and its conditional system state $\rho_S(t)$ still obeys the exact Schr\"odinger equation~\eqref{eq:schrodinger_emergent} everywhere on the support of $w(t)$.
\end{proposition}

\begin{proof}
Applying $\widehat{C}_{univ} = -i\hbar\frac{d}{dt} \otimes I_S + I_C \otimes \widehat{H}_S$ to Eq.~\eqref{eq:wavepacket_state} and integrating by parts gives
\begin{align}
    \widehat{C}_{\mathrm{univ}}}|\Psi_w\rangle &= \int_{-\infty}^{\infty} dt \, |t\rangle_C \otimes \left(-i\hbar \frac{d}{dt} + \widehat{H}_S\right)\!\left[w(t) e^{-i\widehat{H}_S t / \hbar} |\psi_0\rangle_S\right] \nonumber\\
    &= -i\hbar \int_{-\infty}^{\infty} dt \, w'(t) \, |t\rangle_C \otimes e^{-i\widehat{H}_S t / \hbar} |\psi_0\rangle_S.
\end{align}
Taking the H_U-norm yields
$\|\widehat{C}_{\mathrm{univ}}}|\Psi_w\rangle\|^2 = \hbar^2 \int_{-\infty}^\infty |w'(t)|^2 dt = \hbar^2 \|w'\|_{L^2}^2$.
Projecting onto $\langle t|_C$ gives $|\Psi_w(t)\rangle = w(t)$

e^{-i\widehat{H}_S t / \hbar}|\psi_0\rangle_S$, so the raw clock marginal is $M_{\Psi_w}(t) = \|\Psi_w(t)\|^2 = |w(t)|^2$, while the normalized conditional state $|\psi(t)\rangle = |\Psi_w(t)\rangle / |w(t)| = e^{-i\widehat{H}_S t/\hbar}|\psi_0\rangle_S$ evolves unitarily under $\widehat{H}_S$.
\end{proof}

%===
% SECTION V: FINITE-CLOCK LINDBLAD DEPHASING
%===
\section{Finite-Clock Corrections and Intrinsic Lindblad Dephasing}
\label{sec:lindblad}

When the clock subsystem is prepared or read out with a finite energy variance $\Delta E_C < \infty$, the Fourier uncertainty relation imposes a fundamental temporal resolution limit:
\begin{equation}
    \sigma_t \gtrsim \frac{\hbar}{2\Delta E_C}.
    \label{eq:clock_uncertainty}
\end{equation}
Consequently, conditioning on an imperfect clock readout at nominal time t averages the ideal unitary evolution $U(t) = e^{-i\widehat{H}_S t/\hbar}$ against the clock pointer's quadrature resolution kernel $G_{\sigma_t}(\tau)$.

\begin{theorem}[Finite-Clock Double-Commutator Master Equation]
\label{thm:lindblad}
Let the observed conditional system density operator under a Gaussian clock resolution kernel $G_{\sigma_t}(\tau) = (2\pi\sigma_t^2)^{-1/2}\exp(-\tau^2/2\sigma_t^2)$ be given by the convolution channel
\begin{equation}
    \rho_S^{\mathrm{obs}}(t) = \int_{-\infty}^{\infty} d\tau \, G_{\sigma_t}(\tau) \, U(t + \tau) \rho_S(0) U^\dagger(t + \tau).
    \label{eq:gaussian_channel}
\end{equation}
Then:
\begin{enumerate}
    \item To second order in σ_t, the short-time conditional state acquires a double-commutator correction:
    \begin{equation}
        \rho_S^{\mathrm{obs}}(t) = U(t)\left[\rho_S(0) - \frac{\sigma_t^2}{2\hbar^2} [\widehat{H}_S, [\widehat{H}_S, \rho_S(0)]] + \mathcal{O}(\sigma_t^4)\right]U^\dagger(t).

\label{eq:short_time_expansion}
\end{equation}
\item In the coarse-grained Markovian regime where clock phase variance accumulates diffusively as $\sigma_t^2(t) = T_C t$ with characteristic clock correlation scale $T_C \sim \hbar/\Delta E_C$, $\rho_S^{obs}(t)$ satisfies the exact, completely positive Lindblad master equation~\cite{Gambini2004,Milburn1991}:
\begin{equation}
\frac{d\rho_S}{dt} = -\frac{i}{\hbar}[\widehat{H}_S, \rho_S] - \frac{T_C}{2\hbar^2} [\widehat{H}_S, [\widehat{H}_S, \rho_S]] = -\frac{i}{\hbar}[\widehat{H}_S, \rho_S] + \frac{T_C}{\hbar^2}\left(\widehat{H}_S \rho_S \widehat{H}_S - \frac{1}{2}\{\widehat{H}_S^2, \rho_S\}\right).
\label{eq:lindblad_master}
\end{equation}
\item In the energy eigenbasis $\widehat{H}_S |E_j\rangle = E_j |E_j\rangle$, diagonal populations $\rho_{jj}(t) = \rho_{jj}(0)$ are conserved, while off-diagonal coherences decay exponentially as
\begin{equation}
\rho_{jk}^{\mathrm{obs}}(t) = \rho_{jk}(0) \exp\left(-i\omega_{jk} t - \Gamma_{jk} t\right), \qquad \Gamma_{jk} = \frac{T_C}{2\hbar^2}(E_j - E_k)^2 \sim \frac{(E_j - E_k)^2}{2\hbar \Delta E_C},
\label{eq:dephasing_rate}
\end{equation}
where $\omega_{jk} = (E_j - E_k)/\hbar$.
\end{enumerate}
\end{theorem}

\begin{proof}
Factoring $U(t+\tau) = U(t)U(\tau)$ in Eq.~\eqref{eq:gaussian_channel} and expanding $U(\tau)\rho_S(0)U^\dagger(\tau)$ via the Baker--Campbell--Hausdorff series in τ gives
\begin{equation}
U(\tau)\rho_S(0)U^\dagger(\tau) = \rho_S(0) + \frac{\tau}{i\hbar}[\widehat{H}_S, \rho_S(0)] - \frac{\tau^2}{2\hbar^2}[\widehat{H}_S, [\widehat{H}_S, \rho_S(0)]] + \mathcal{O}(\tau^3).
\end{equation}
Integrating against $G_{\sigma_t}(\tau)$ using $\int d\tau G_{\sigma_t}(\tau)\tau = 0$ and $\int d\tau G_{\sigma_t}(\tau)\tau^2 = \sigma_t^2$ yields Eq.~\eqref{eq:short_time_expansion}.

To obtain the exact behavior in the energy eigenbasis, expand $\rho_S(0) = \sum_{j,k} \rho_{jk}(0)|E_j\rangle\langle E_k|$ in Eq.~\eqref{eq:gaussian_channel}:

\begin{equation}
    \rho_{jk}^{\mathrm{obs}}(t) = \rho_{jk}(0) e^{-i\omega_{jk} t}
\int_{-\infty}^{\infty} \frac{d\tau}{\sqrt{2\pi}\sigma_t}
\exp\left(-\frac{\tau^2}{2\sigma_t^2} - i\omega_{jk}\tau\right) =
\rho_{jk}(0) \exp\left(-i\omega_{jk} t - \frac{1}{2}\omega_{jk}^2
\sigma_t^2\right).
    \label{eq:exact_gaussian_fourier}
\end{equation}
Substituting the diffusive clock variance scaling $\sigma_t^2(t) = T_C t$ into
Eq.~\eqref{eq:exact_gaussian_fourier} produces
Eq.~\eqref{eq:dephasing_rate}. Differentiating
Eq.~\eqref{eq:dephasing_rate} with respect to t gives
\begin{equation}
    \frac{d\rho_{jk}^{\mathrm{obs}}}{dt} = \left[-i\omega_{jk} -
\frac{T_C}{2}\omega_{jk}^2\right] \rho_{jk}^{\mathrm{obs}}(t).
\end{equation}
Using $[\widehat{H}_S, |E_j\rangle\langle E_k|] =
\hbar\omega_{jk}|E_j\rangle\langle E_k|$ and $[\widehat{H}_S,
[\widehat{H}_S, |E_j\rangle\langle E_k|]] = \hbar^2\omega_{jk}^2
|E_j\rangle\langle E_k|$, along with the algebraic identity
$-\frac{1}{2}[\widehat{H}_S,[\widehat{H}_S,\rho_S]] = \widehat{H}_S\rho_S\widehat{H}_S - \frac{1}{2}\{\widehat{H}_S^2,\rho_S\}$, establishes
Eq.~\eqref{eq:lindblad_master} in standard
Gorini--Kossakowski--Sudarshan--Lindblad (GKSL) form with a single
Hermitian jump operator $L = \sqrt{T_C/\hbar^2}\widehat{H}_S$.
\end{proof}

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% SECTION VI: MODULAR FLOW & WIGNER PHASE-SPACE QUADRATURE
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\section{Equivalence to Modular Flow and Wigner Phase-Space
Quadrature}
\label{sec:modular_wigner}

\subsection{Tomita--Takesaki Modular Flow as State-Generated
Quadrature Time}
\begin{proposition}[Modular Time Equivalence]
\label{prop:modular}
Let $|\Psi\rangle = \sum_k \alpha_k |n_k\rangle_C \otimes
|E_k\rangle_S$ be a globally stationary bipartite state with
non-vanishing Schmidt amplitudes $\alpha_k > 0$. Then the reduced states $\rho_C
= \operatorname{Tr}_S(|\Psi\rangle\langle\Psi|)$ and $\rho_S =
\operatorname{Tr}_C(|\Psi\rangle\langle\Psi|)$ define a

Tomita--Takesaki modular operator $\Delta_\Psi = \rho_C \otimes \rho_S^{-1}$ on the support algebra $A_S = B(H_S)$. When the clock-system entanglement spectrum follows a Boltzmann distribution $|\alpha_k|^2 = Z^{-1} e^{-\beta E_k}$, the modular automorphism group $\sigma_s^\Psi(A_S) = \Delta_\Psi^{is}(I_C \otimes A_S)\Delta_\Psi^{-is}$ coincides identically with Heisenberg evolution under Quadrature Time with $t = -\hbar\beta s$.
\end{proposition}

\begin{proof}
Tracing out H_C from Eq.~\eqref{eq:entangled_stationary} gives the faithful reduced system state $\rho_S = \sum_k |\alpha_k|^2 |E_k\rangle\langle E_k|_S$, whose modular Hamiltonian is $K_S = -\ln \rho_S = -\sum_k (\ln|\alpha_k|^2)|E_k\rangle\langle E_k|_S$. Substituting $|\alpha_k|^2 = Z^{-1} e^{-\beta E_k}$ gives $K_S = \beta \widehat{H}_S + (\ln Z) I_S$. Consequently, for any observable $A_S \in A_S$,
\begin{equation}
    \sigma_s^\Psi(A_S) = \rho_S^{is} A_S \rho_S^{-is} = e^{-is K_S} A_S e^{is K_S} = e^{-is\beta \widehat{H}_S} A_S e^{is\beta \widehat{H}_S}.
\end{equation}
Comparing with the conditional Heisenberg evolution $A_S(t) = e^{i\widehat{H}_S t/\hbar} A_S e^{-i\widehat{H}_S t/\hbar}$ induced by Theorem~\ref{thm:schrodinger} establishes exact equivalence under $t = -\hbar\beta s$.
\end{proof}

\subsection{Wigner--Weyl Phase-Space Quadrature (``Squaring the Circle'')}
For a continuous n-dimensional system $H_S = L^2(R^n, d^n q)$, let $\psi(q, \theta) = ((\theta|_C \otimes \langle q|_S)|\Psi\rangle$ be the conditional configuration-space wave function at clock angle $\theta \in S^1$. The conditional Wigner distribution~\cite{Wigner1932} maps the Hilbert-space state quadratically onto the Cartesian phase space $(q, p) \in R^{2n}$:
\begin{equation}
    W_\Psi(q, p; \theta) := \frac{1}{(2\pi\hbar)^n} \int_{\mathbb{R}^n} d^n y \, e^{-i p \cdot y / \hbar} \, \psi\!\left(q + \frac{y}{2}, \theta\right) \psi^*\!\left(q - \frac{y}{2}, \theta\right).
    \label{eq:wigner_quadrature}
\end{equation}
Integrating $W_\Psi(q, p; \theta)$ over all phase space $(q, p) \in R^{2n}$ recovers the clock

quadrature density $2\pi M_{\Psi}(\theta) = \int d^n q d^n p W_{\Psi}(q,p;\theta)$, while differentiating with respect to the clock angle θ via Theorem~\ref{thm:schrodinger} yields the relational Moyal equation:

```
\begin{equation}
    \omega \frac{\partial}{\partial \theta} W_\Psi(q, p; \theta) =
\{H_S(q, p), W_\Psi(q, p; \theta)\}_{\mathrm{Moyal}}
\xrightarrow{\hbar \to 0} \{H_S, W_\Psi\}_{\mathrm{Poisson}}.
    \label{eq:moyal_relational}
\end{equation}
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Thus, $|\Psi\rangle$ simultaneously squares the circular clock fiber S_C^1 into the temporal measure $M_{\Psi}(\theta)$ and squares the system Hilbert space into the symplectic phase-space arena R^{2n}.

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% SECTION VII: CIRCUIT-QED EXPERIMENTAL PROTOCOL
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\section{Experimental Protocol in Superconducting Circuit QED}
\label{sec:experiment}
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Whereas photonic polarization experiments have demonstrated the ideal kinematic Page--Wootters mechanism in two-qubit systems~\cite{Moreva2014}, testing the Hilbert-Space Quadrature Conjecture requires measuring the finite-clock Lindblad dephasing rate Γ_{jk} (Theorem~\ref{thm:lindblad}) as an explicit function of the clock's quantum energy spread ΔE_C. We propose a superconducting circuit-QED realization:

\begin{enumerate}

\item \textbf{Bipartite Hardware ($H_C \otimes H_S$):} Couple a frequency-tunable transmon qubit ($H_S = C^2$, transition frequency $\omega_s/2\pi \approx 5.0 GHz$) dispersively to a high-Q 3D microwave cavity or coplanar waveguide resonator (H_C, frequency $\omega_c/2\pi \approx 6.0 GHz$, photon number operator $\widehat{N} = a^\dagger a$). In the dispersive regime, the interaction Hamiltonian is

```
    \begin{equation}
        \widehat{H}_{\mathrm{int}} = \hbar \chi \, a^\dagger a \,
|1\rangle\langle 1|_S,
        \label{eq:dispersive_H}
    \end{equation}
```

    with dispersive shift $\chi/2\pi \approx -1.0 MHz$. Here, the resonator acts as

the quantum clock whose phase rotation tracks the relational evolution of the transmon with effective system energy splitting $E_1 - E_0 = \hbar\chi_{eff}$.

\item \textbf{Clock Energy Spread Preparation:} Displace the resonator clock into a coherent state $\vert{}\alpha\rangle_C = e^{-\vert{}\alpha\vert{}^2/2}\sum_{n=0}^\infty \frac{\alpha^n}{\sqrt{n!}}\vert{}n\rangle_C$ with tunable mean photon number $\bar{n} = \vert{}\alpha\vert{}^2 \in [10, 10^4]$ and Poissonian number standard deviation $\Delta n = \sqrt{\bar{n}}$, fixing the clock energy spread at

\begin{equation}
\Delta E_C = \hbar \omega_c \sqrt{\bar{n}}.
\end{equation}

Simultaneously prepare the transmon in the equatorial superposition $\vert{}\psi_0\rangle_S = \frac{1}{\sqrt{2}}(\vert{}0\rangle + \vert{}1\rangle)_S$ via a $\pi/2$ pulse.

\item \textbf{Quadrature Readout and Scaling Law:} Perform phase-sensitive heterodyne detection on the cavity leakage field (implementing the covariant phase POVM $E(d\theta)$ on S^1) conditioned alongside dispersive state tomography of the transmon's off-diagonal coherence $\vert{}\rho_{01}(\theta)\vert{}$. By Theorem~\ref{thm:lindblad}, the intrinsic clock-induced dephasing rate scales inversely with $\sqrt{\bar{n}}$:

\begin{equation}
\Gamma_{01}^{\mathrm{clock}}(\bar{n}) = \frac{(E_1 - E_0)^2}{2\hbar \Delta E_C} \propto \frac{\chi^2}{2\omega_c \sqrt{\bar{n}}}.
\label{eq:circuit_qed_scaling}
\end{equation}

Sweeping $\bar{n}$ across three orders of magnitude while holding cavity temperature and intrinsic qubit T_2^* constant isolates the $\bar{n}^{-1/2}$ quadrature dephasing signature from standard thermal photon shot noise (which increases with $\bar{n}$).
\end{enumerate}

%===
% SECTION VIII: CONCLUSION
%===
\section{Conclusion}
\label{sec:conclusion}

By separating the universal cover constraint

$\widehat{C}_{\mathrm{univ}}|\Psi\rangle = 0$ on $L^2(R)$ from the covariant Holevo phase POVM $E(d\theta)$ on the Hardy space $H^2(S^1)$ via the Bloch--Floquet winding quotient Π_ω, the Hilbert-Space Quadrature Conjecture provides a mathematically self-consistent foundation for emergent quantum time. Standard unitary Schr\"odinger dynamics emerges conditionally in the infinite-bandwidth clock limit (Theorem~\ref{thm:schrodinger}), while physical clocks with finite energy spread ΔE_C obey a completely positive double-commutator Lindblad master equation (Theorem~\ref{thm:lindblad}) that is directly testable in contemporary circuit-QED architectures.

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% SECTION IX: RELATIVISTIC SPACETIME EXTENSION
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\section{Relativistic Spacetime Extension: Klein--Gordon, Dirac, and
Proper-Time Quadrature}
\label{sec:relativistic}

\subsection{The Klein--Gordon Mass-Shell Constraint and Hardy-Space
Selection}
In $(1+3)$-dimensional Minkowski spacetime with metric
$\eta_{\mu\nu} = diag(+1,-1,-1,-1)$, let the universal kinematical Hilbert space be
$H_U^{kin} = L^2(R, dx^0) \otimes L^2(R^3, d^3x) \otimes H_{int}$, with temporal momentum $\widehat{P}_0 = -\frac{1}{c}\widehat{P}_t$, spatial
momentum $\widehat{P} = -i\hbar\nabla$, and mass operator $\widehat{M} = mI_{int} + \frac{1}{c^2}\widehat{H}_{int}$. Relativistic
reparameterization invariance imposes the quadratic Klein--Gordon
mass-shell constraint:
\begin{equation}
    \widehat{C}_{\mathrm{KG}}|\Psi\rangle = \left(\widehat{P}_\mu
\widehat{P}^\mu - \widehat{M}^2 c^2\right)|\Psi\rangle =
\frac{1}{c^2}\widehat{C}_+ \widehat{C}_- |\Psi\rangle = 0, \qquad
\widehat{C}_\pm := \widehat{P}_t \otimes I_S \pm I_C \otimes
\widehat{H}_{\mathrm{rel}},
    \label{eq:kg_constraint}
\end{equation}
where $\widehat{H}_{rel} := +\sqrt{c^2\widehat{P}^2 + \widehat{M}^2 c^4} > 0$.

\begin{theorem}[Hardy-Space Selection of Positive Frequency and
Positive-Definite Klein--Gordon Quadrature]
\label{thm:kg_hardy}
Under the Bloch--Floquet winding quotient $\Pi_\omega : L^2(R) \to L^2(S^1)$ followed by
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the orthogonal Hardy-space projection $P_+ : L^2(S^1) \to H^2(S^1)$ onto lower-bounded clock states ($\operatorname{Spec}(\widehat{N}) = \mathbb{N}_0$), the negative-frequency mass-shell sheet $\ker(\widehat{C}_-)$ is annihilated identically: $(P_+ \otimes I_S)\ker(\widehat{C}_-) = \{0\}$. The surviving physical subspace $H_+ = \ker(\hbar\omega\widehat{N} \otimes I_S - I_C \otimes \widehat{H}_{rel})$ yields a conditional wave function $|\psi(\theta)\rangle = ((\theta|_C \otimes I_S)|\Psi_+\rangle$ obeying the positive-frequency relativistic Schr\"odinger--Foldy equation ($t = \theta/\omega$),

\begin{equation}
    i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle = \sqrt{c^2\widehat{\mathbf{P}}^2 + \widehat{M}^2 c^4}\,|\psi(t)\rangle,
\end{equation}

with a strictly nonnegative, conserved conditional spatial quadrature density $\rho_{\mathrm{KG}}(\mathbf{x}|\theta) = |\langle\mathbf{x}|\psi(\theta)\rangle|^2 / \||\psi(\theta)\|^2 \ge 0$.
\end{theorem}

\begin{proof}
In the Fourier-momentum basis $\{\vert{}n\rangle_C \otimes \vert{}\mathbf{p},k\rangle_S\}$, the factored constraint $\widehat{C}_\pm\vert{}\Psi_\pm\rangle = 0$ enforces $\hbar\omega n = \pm E_{p,k}$ with $E_{p,k} = \sqrt{c^2p^2 + M_k^2c^4} > 0$. Because the negative-frequency sheet requires $n = -E_{p,k}/(\hbar\omega) < 0$, the Hardy projector $\mathbb{P}_+ = \sum_{n=0}^\infty \vert{}n\rangle\langle n\vert{}_C$ annihilates $\vert{}\Psi_-\rangle$, isolating $\vert{}\Psi_+\rangle$ and rendering the conditional spatial probability density a positive-definite Hilbert-space norm square on H_S.
\end{proof}

\subsection{The Dirac Constraint and Finite-Clock Damping of Zitterbewegung}

For spin-$\frac{1}{2}$ fermions on $H_S = L^2(R^3) \otimes C^4$, multiplying the covariant Dirac constraint $(\gamma^\mu \widehat{P}_\mu - mc I_4)|\Psi\rangle = 0$ by $c\gamma^0$ yields

\begin{equation}
    \widehat{C}_{\mathrm{Dirac}}|\Psi\rangle = \left(\widehat{P}_t \otimes I_S + I_C \otimes \widehat{H}_D\right)|\Psi\rangle = 0, \qquad \widehat{H}_D = c\,\boldsymbol{\alpha}\cdot\widehat{\mathbf{P}} + \beta m c^2.
    \label{eq:dirac_constraint}
\end{equation}

By Theorem~\ref{thm:lindblad}, for a clock with finite energy spread ΔE_C, coherent *Zitterbewegung* oscillations between positive- and

negative-energy Dirac spinor branches ($E_+ - E_- = 2E_p \geq 2mc^2$) experience intrinsic Lindblad dephasing at the rate

$$\Gamma_{\mathrm{Zitt}}(\mathbf{p}) = \frac{(2E_{\mathbf{p}})^2}{2\hbar \Delta E_C} = \frac{2(c^2\mathbf{p}^2 + m^2c^4)}{\hbar \Delta E_C} \ge \frac{2m^2c^4}{\hbar \Delta E_C}, \label{eq:zitt_damping}$$

dynamically projecting the conditional evolution onto the smooth Foldy--Wouthuysen positive-energy sector whenever $\Delta E_C \ll 2mc^2$.

Internal Proper-Time Quadrature and Quantum Lorentz Dilation

When an internal Hardy clock $H_{C,int} = H^2(S^1)$ with Hamiltonian $\widehat{H}_{int} = \hbar\omega_0 \widehat{N}_{int}$ is carried by a moving particle of rest mass m, mass-energy equivalence $\widehat{M} = mI + \widehat{H}_{int}/c^2$ yields the first-order expansion

$$\widehat{H}_{\mathrm{rel}} = \widehat{E}_{\mathbf{P}} \otimes I_{\mathrm{int}} + \widehat{\gamma}(\widehat{\mathbf{P}})^{-1} \otimes \hbar\omega_0 \widehat{N}_{\mathrm{int}} + \mathcal{O}\!\left(\frac{\widehat{H}_{\mathrm{int}}^2}{mc^2}\right), \qquad \widehat{\gamma}(\widehat{\mathbf{P}}) = \sqrt{1 + \frac{\widehat{\mathbf{P}}^2}{m^2 c^2}}. \label{eq:proper_time_hamiltonian}$$

Conditioning on the laboratory reference clock t and reading out the internal proper-time POVM $E_{int}(d\tau)$ shifts the conditional proper-time distribution to $\langle \tau(t) \rangle = \langle \widehat{\gamma}^{-1} \rangle t$, while a center-of-mass momentum superposition $c_1|\mathbf{p}_1\rangle + c_2|\mathbf{p}_2\rangle$ modulates the internal clock coherence by the quantum twin-paradox visibility factor $\sqrt{|c_1|^4 + |c_2|^4 + 2|c_1|^2|c_2|^2 \cos[(n-m)\omega_0(\gamma_1^{-1} - \gamma_2^{-1})t]}$ ~\cite{SmithAhmadi2020}.

Stationary Curved Spacetime: Schwarzschild Redshift, Twin-Shell Decoherence, and Horizon Dephasing

\label{subsec:curved_spacetime}

Let $(M, g_{\mu\nu})$ be a static, globally hyperbolic spacetime with timelike Killing vector $\xi^\mu = (\partial_{x^0})^\mu$ ($x^0 = ct$) and line element

$$ds^2 = N(\mathbf{x})^2 c^2 dt^2 - h_{ij}(\mathbf{x})\,dx^i dx^j,$$

\qquad N(\mathbf{x}) := \sqrt{g_{00}(\mathbf{x})},
    \label{eq:static_metric}
\end{equation}
which for the exterior Schwarzschild geometry ($r > r_s := 2GM_G/c^2$) has lapse function $N(r) = \sqrt{1 - r_s/r}$. Multiplying the covariant Klein--Gordon mass-shell constraint by $c^2 N(x)^2$ and applying the Bloch--Floquet winding quotient Π_{ω_∞} and Hardy projection P_+ onto an asymptotic reference clock $H_{C,\infty} = H^2(S^1)$ at spatial infinity ($N(\infty) = 1$) yields the positive-frequency curved-spacetime constraint
\begin{equation}
    \widehat{C}_+^{\mathrm{grav}}|\Psi_+\rangle =
\left(\hbar\omega_\infty \widehat{N}_\infty \otimes I_S - I_C \otimes
\widehat{H}_{\mathrm{grav}}\right)|\Psi_+\rangle = 0, \qquad
\widehat{H}_{\mathrm{grav}} =
\sqrt{N(\widehat{\mathbf{X}})}{\left(c^2\widehat{\mathbf{P}}_\Sigma^2
+ \widehat{M}^2 c^4\right)}N(\widehat{\mathbf{X}}).
    \label{eq:grav_constraint}
\end{equation}

\begin{theorem}[Emergent Gravitational Time Dilation and Twin-Shell Quadrature Decoherence]
\label{thm:grav_dilation}
Consider a quantum probe of rest mass m carrying an internal circular Hardy clock $H_{C,int} = H^2(S^1)$ with internal Hamiltonian $\widehat{H}_{int} = \hbar\omega_0 \widehat{N}_{int}$, total invariant mass $\widehat{M} = mI_{int} + \widehat{H}_{int}/c^2$, and Holevo proper-time POVM $E_{int}(d\tau)$. In the quasi-static regime ($\langle
c^2\widehat{\mathbf{P}}_\Sigma^2\rangle \ll m^2c^4$,
$\hbar\omega_0\langle\widehat{N}_{\mathrm{int}}\rangle \ll mc^2$), the Killing Hamiltonian expands as $\widehat{H}_{grav} \approx \widehat{H}_{cm}^{grav} \otimes I_{int} + N(\widehat{X}) \otimes \hbar\omega_0 \widehat{N}_{int}$. Conditioning on the asymptotic clock reading $t = \theta_\infty/\omega_\infty$ yields:
\begin{enumerate}
    \item \textbf{Schwarzschild Proper-Time Shift:} For a probe localized at radius $r > r_s$, the conditional proper-time quadrature distribution shifts according to the exact Tolman--Ehrenfest redshift relation:
    \begin{equation}
        Q_\Psi(\tau \mid t, r) = Q_\Psi\!\left(\tau - \sqrt{1 - \frac{r_s}{r}}\,t \,\middle|\, 0, r\right), \qquad \langle \tau(t) \rangle_r = \sqrt{1 - \frac{r_s}{r}}\,t.
        \label{eq:schwarzschild_shift}
    \end{equation}
    \item \textbf{Twin-Shell Gravitational Decoherence:} If the

probe's center of mass is prepared in a coherent spatial superposition $\frac{1}{\sqrt{2}}(\vert{}r_1\rangle + \vert{}r_2\rangle)$ across two radial shells with redshift differential $\Delta N_{12} = \sqrt{1 - r_s/r_2} - \sqrt{1 - r_s/r_1}$ and internal Gaussian energy spread $\Delta E_{int} = \hbar\omega_0 \Delta n_{int}$, tracing out $H_{C,int}$ suppresses the spatial interferometric visibility as~\cite{Zych2011,Pikovski2015}:

\begin{equation}
\mathcal{V}_{\mathrm{cm}}(t) = 2|\langle r_1 | \rho_{\mathrm{cm}}(t) | r_2 \rangle| = \exp\!\left(-\frac{(\Delta E_{\mathrm{int}})^2 (\Delta N_{12})^2 t^2}{2\hbar^2}\right) = \exp\!\left(-\frac{t^2}{\tau_{\mathrm{grav}}^2}\right), \qquad \tau_{\mathrm{grav}} = \frac{\sqrt{2}\,\hbar}{\Delta E_{\mathrm{int}}|\Delta N_{12}|}.
\label{eq:twin_shell_visibility}
\end{equation}

\item \textbf{Non-Local Lindblad Horizon Divergence:} When a clock at radius $r_C > r_s$ with local proper energy spread ΔE_C^{loc} parameterizes the evolution of a system at radius $r_S > r_s$, the coordinate-time Lindblad dephasing rate of Theorem~\ref{thm:lindblad} becomes

\begin{equation}
\Gamma_{jk}^{\mathrm{grav}}(r_S, r_C) = \frac{N(r_S)^2}{N(r_C)}\frac{(E_j^{\mathrm{loc}} - E_k^{\mathrm{loc}})^2}{2\hbar \Delta E_C^{\mathrm{loc}}} = \frac{1 - r_s/r_S}{\sqrt{1 - r_s/r_C}}\,\Gamma_{jk}^{(0)}.
\label{eq:horizon_lindblad}
\end{equation}

For co-located systems ($r_S = r_C = r$), transforming to local proper time $d\tau = \sqrt{1 - r_s/r}dt$ cancels $N(r)$ identically, preserving the Einstein Equivalence Principle; whereas for a near-horizon clock ($r_C \to r_s^+$) timing a distant system ($r_S > r_s$), $\Gamma_{jk}^{grav} \propto (1 - r_s/r_C)^{-1/2} \to \infty$~\cite{CastroRuiz2017}.

\end{enumerate}
\end{theorem}

\begin{proof}

Conditioning on asymptotic time t generates the unitary flow $e^{-i\widehat{H}_{grav}t/\hbar}$. On a single radial shell $\vert{}r\rangle$, the internal clock state $\vert{}n\rangle_{\mathrm{int}}$ acquires the redshifted phase $e^{-in\omega_0 N(r)t}$, so projecting onto the Holevo bra $(\theta_{\mathrm{int}}\vert{}_{\mathrm{int}} = \sum_{n=0}^\infty e^{-in\omega_0\tau}\langle n\vert{}_{\mathrm{int}}$ shifts the argument of the quadrature density from τ to $\tau - N(r)t$, proving Eq.~\eqref{eq:schwarzschild_shift}. For the superposition

$\frac{1}{\sqrt{2}}(\vert{}r_1\rangle + \vert{}r_2\rangle)\otimes\sum_n c_n\vert{}n\rangle_{\mathrm{int}}$, tracing out $H_{C,int}$ yields $\langle r_1\vert{}\rho_{\mathrm{cm}}(t)\vert{}r_2\rangle = \frac{1}{2}e^{imc^2\Delta N_{12}t/\hbar}\sum_{n=0}^\infty \vert{}c_n\vert{}^2 e^{in\omega_0\Delta N_{12}t}$, whose Gaussian Fourier transform over $\vert{}c_n\vert{}^2$ gives Eq.~\eqref{eq:twin_shell_visibility}. Finally, substituting the redshifted Killing energy gaps $\Delta E_{jk}^{Killing} = N(r_S)(E_j^{loc} - E_k^{loc})$ and the redshifted clock correlation scale $T_C^{Killing} = \hbar/[N(r_C)\Delta E_C^{loc}]$ into Eq.~\eqref{eq:dephasing_rate} yields Eq.~\eqref{eq:horizon_lindblad}. \end{proof}

\subsection{Cosmological Minisuperspace: Scalar Field as a Quantum Clock for the Scale Factor}
\label{subsec:minisuperspace}

In canonical quantum cosmology, the Wheeler--DeWitt equation for a flat Friedmann--Robertson--Walker (FRW) universe sourced by a homogeneous scalar field φ with potential $V(\varphi)$ is given by~\cite{DeWitt1967,Kuchar1992}
\begin{equation}
    \widehat{H}_{\mathrm{WDW}} |\Psi\rangle = \left[\frac{1}{2m_0} a^{-p} \frac{\partial}{\partial a} \left(a^p \frac{\partial}{\partial a} \right) - \frac{1}{2a^3} \frac{\partial^2}{\partial \varphi^2} + a^3 V(\varphi) \right] |\Psi(a, \varphi)\rangle = 0,
    \label{eq:wdw_frw_app}
\end{equation}
where $m_0 = 3\pi/(2G)$. Treating the homogeneous scalar field φ as the internal quantum clock sector ($H_C = L^2(R)$) and the scale factor a as the system sector (H_S), conditioning on a scalar field clock reading via the quadrature map $E(d\varphi) = |\varphi\rangle\langle\varphi|\,d\varphi$ yields the conditional scale factor state $|\psi(a|\varphi)\rangle = \langle\varphi|\Psi\rangle$. Under the adiabatic Born--Oppenheimer expansion, $|\psi(a|\varphi)\rangle$ satisfies the emergent Tomonaga--Schwinger cosmological Schr\"odinger equation:
\begin{equation}
    i \frac{\partial}{\partial \varphi} |\psi(a | \varphi)\rangle = \widehat{H}_{\mathrm{cosmo}}(a, \varphi) |\psi(a | \varphi)\rangle, \qquad \widehat{H}_{\mathrm{cosmo}} = \sqrt{\frac{2m_0}{a^3}} \left(a^3 V(\varphi) - \frac{1}{2m_0} a^{-p}\partial_a(a^p \partial_a)\right).
    \label{eq:cosmo_schrodinger_app}
\end{equation}
Furthermore, finite scalar-field clock uncertainty $\Delta\varphi$ induces quantum

gravitational backreaction governed by Theorem~\ref{thm:lindblad}, predicting an intrinsic Hubble expansion variance $\sigma_H^2 = (\partial_\varphi H)^2 (\Delta\varphi)^2 + \gamma_{cosmo}/(\Delta\varphi)^2$ that dynamically suppresses macroscopic geometry superpositions and drives emergent classical spacetime.

\subsection{The Hartle--Hawking No-Boundary Wave Function via Covariant Phase POVMs}
\label{subsec:hartle_hawking}

In quantum cosmology, the Hartle--Hawking no-boundary wave function is defined by the Euclidean path integral over compact 4-geometries g and fields ϕ bounded only by spatial data (h_{ij}, φ)~\cite{HartleHawking1983}:
\begin{equation}
    \Psi_{\mathrm{HH}}[h_{ij}, \varphi] = \int_{\partial g = (h, \varphi)} \mathcal{D}g \, \mathcal{D}\phi \, \exp\left(-I_E[g, \phi]\right).
    \label{eq:hh_path_app}
\end{equation}
Within the Hilbert-Space Quadrature Conjecture, Euclidean proper time is the circular phase parameter $\theta_E \in S_E^1$ of an internal Euclidean clock sector $H_{C,E} = L^2(S_E^1)$, and the no-boundary wave function is a state-generated quadrature projection.

\begin{theorem}[Quadrature Formulation of the Hartle--Hawking State]
\label{thm:hartle_hawking}
Let $H_U = H_{C,E} \otimes H_{geom}$ be the universe comprising the Euclidean clock circle and gravitational superspace. The no-boundary state $|\Psi_{\mathrm{NB}}\rangle$ is generated by integrating the covariant Euclidean phase POVM $E_E(d\theta_E) = \frac{1}{2\pi}|\theta_E)(\theta_E|d\theta_E$ over S_E^1 anchored at a smooth regular cap:
\begin{equation}
    |\Psi_{\mathrm{NB}}\rangle = \int_0^{2\pi} \frac{d\theta_E}{2\pi} \, |\theta_E)_C \otimes \left(\int_{\mathrm{cap}}^{\theta_E} \mathcal{D}g\,\mathcal{D}\phi \, e^{-I_E[g,\phi]} |\mathrm{geom}(\theta_E)\rangle \right),
    \label{eq:nb_state_app}
\end{equation}
such that $\Psi_{\mathrm{HH}}[h_{ij}, \varphi; \theta_E] = \langle \theta_E, h_{ij}, \varphi | \Psi_{\mathrm{NB}} \rangle$. The transition from Euclidean no-boundary quantum cosmology to Lorentzian real-time expansion corresponds operationally to the analytic unwrapping of the Euclidean clock phase POVM into the universal cover group-averaged constraint $P_{phys} = \delta(\widehat{C}_{univ})$ of Section~\ref{sec:universal_cover}.

\end{theorem}

\subsection{Near-Horizon Limit ($r_C \to r_s^+$), Bisognano--Wichmann Theorem, and Hawking--Unruh Temperature}
\label{subsec:near_horizon}

Near the Schwarzschild event horizon $r = r_s + \rho$ ($\rho \ll r_s$), the lapse function is $N(r) \approx \kappa\rho/c^2$ with surface gravity $\kappa = c^4/(4GM_G)$, reducing the metric to Rindler spacetime. By the Bisognano--Wichmann theorem, the modular operator Δ_Ψ of Proposition~\ref{prop:modular} is generated by the Rindler boost operator K.

\begin{theorem}[Near-Horizon Modular Flow and Hawking--Unruh Temperature]
\label{thm:near_horizon}
Restricting the universal state $|\Psi\rangle$ to the near-horizon Rindler wedge yields a modular KMS thermal state with period $\Delta s = 2\pi$. Identifying modular flow with Quadrature Time via $t = -\hbar\beta s$ yields the Hawking--Unruh temperature:
\begin{equation}
    T_H = \frac{\hbar \kappa}{2\pi k_B c} = \frac{\hbar c^3}{8\pi G M_G k_B}.
    \label{eq:hawking_unruh}
\end{equation}
Simultaneously, the near-horizon limit of the gravitational Lindblad rate (Eq.~\eqref{eq:horizon_lindblad}) diverges as $\Gamma_{jk}^{grav} \propto \rho^{-1/2} \to \infty$ as $\rho \to 0$, reflecting infinite thermal blueshifting of internal clock fluctuations at the event horizon.
\end{theorem}

%==
% APPENDIX A: PYTHON NUMERICAL MODEL
%==
\appendix
\section{Discrete Numerical Model and Python Implementation ($C^N \otimes C^2$)}
\label{app:python}

To illustrate the convergence of the discrete Pegg--Barnett clock $H_C^{(N)} = C^N$ to continuous Schr\"odinger evolution as $N \to \infty$, as well as the exact finite-clock dephasing envelope $|\rho_{01}^{finite}(\theta)| = \frac{1}{2}\exp(-\omega_{ratio}^2\sigma_\theta^2/2)$ where $\sigma_\theta = (2\Delta n)^{-1}$, the listing below provides a self-contained Python

simulation of the bipartite state $|\Psi\rangle \in \mathbb{C}^N \otimes \mathbb{C}^2$.

\begin{lstlisting}[style=pythonstyle, caption={Self-contained Python simulation of the discrete $C^N \otimes C^2$ Quadrature Time model, demonstrating exact bipartite conditioning, finite-clock Lindblad dephasing, and periodic recurrence on S^1.}]

```
import numpy as np
import matplotlib.pyplot as plt

def exact_bipartite_quadrature(N=128, Delta_n=8.0, omega_ratio=1.0):
    """
    Computes the exact conditional quadrature dynamics directly from a
    bipartite entangled state |Psi> in C^N (clock) x C^2 (system)
    using a covariant smeared clock POVM of energy spread Delta_n.
    """
    theta = 2.0 * np.pi * np.arange(N) / N
    n = np.arange(N)
    n0 = N // 2

    # 1. Clock Readout POVM envelope with energy spread Delta_n
    #    Intrinsic phase uncertainty: sigma_theta = 1 / (2 * Delta_n)
    env_n = np.exp(-(n - n0)**2 / (4.0 * max(Delta_n, 1e-6)**2))
    env_n /= np.linalg.norm(env_n)

    # 2. Ideal conditional qubit history states |psi_ideal(theta_k)>
    #    |psi(theta)> = (|0> + exp(-i * omega_ratio * theta)|1>) /
sqrt(2)
    psi_ideal_theta = np.zeros((N, 2), dtype=complex)
    psi_ideal_theta[:, 0] = 1.0 / np.sqrt(2.0)
    psi_ideal_theta[:, 1] = np.exp(-1j * omega_ratio * theta) /
np.sqrt(2.0)

    # 3. Intrinsic Clock Pointer Quadrature Profile G(theta - theta')
    pointer_amp = np.fft.ifft(env_n) * np.sqrt(N)
    G_profile = np.abs(pointer_amp)**2
    G_profile /= G_profile.sum()

    # 4. Exact conditional density matrix via cyclic convolution on
S^1
    rho_ideal = np.einsum("ki,kj->kij", psi_ideal_theta,
np.conj(psi_ideal_theta))
    rho_finite = np.zeros_like(rho_ideal)
    for k_idx in range(N):
        weights = np.roll(G_profile, k_idx)
        rho_finite[k_idx] = np.einsum("l,lij->ij", weights, rho_ideal)
```

```
    # 5. Analytical Gaussian / Lindblad dephasing prediction
    sigma_theta = 1.0 / (2.0 * Delta_n)
    analytical_coh = 0.5 * np.exp(-0.5 * (omega_ratio *
sigma_theta)**2)

    return theta, G_profile, rho_ideal, rho_finite, sigma_theta,
analytical_coh


if __name__ == "__main__":
    N = 128
    Delta_n = 4.0
    omega_ratio = 2.0

    theta, G_prof, rho_id, rho_fin, sig_th, ana_coh =
exact_bipartite_quadrature(
        N=N, Delta_n=Delta_n, omega_ratio=omega_ratio
    )

    fig, axs = plt.subplots(1, 2, figsize=(12, 4.5))

    # Panel 1: Clock Pointer Quadrature Profile on S^1
    axs[0].plot(theta, np.fft.fftshift(G_prof), "o-", markersize=3,
color="#1f77b4")
    axs[0].set_xlabel(r"Clock Phase Angle θ ∈ [0,2π)")
    axs[0].set_ylabel(r"Pointer Quadrature G_σθ(θ − π)")
    axs[0].set_title(rf"Clock Quadrature Kernel (N = N,Δn = Delta_n)")
    axs[0].grid(True, alpha=0.3)

    # Panel 2: Real Part and Magnitude of Qubit Coherence
rho_01(theta)
    axs[1].plot(theta, np.real(rho_id[:, 0, 1]), "--", color="gray",
alpha=0.7,
                label=r"Ideal Re[ρ01(θ)]")
    axs[1].plot(theta, np.real(rho_fin[:, 0, 1]), "-",
color="#d62728", lw=2,
                label=r"Finite-Clock Re[ρ01^obs(θ)]")
    axs[1].axhline(ana_coh, color="#2ca02c", ls=":", lw=1.8,
                   label=rf"Analytical Envelope |ρ01| = ana_coh:.4f")
    axs[1].axhline(-ana_coh, color="#2ca02c", ls=":", lw=1.8)
    axs[1].set_xlabel(r"Emergent Clock Phase θ = ωt")
    axs[1].set_ylabel(r"Conditional Coherence ρ01(θ)")
    axs[1].set_title("Conditional Schr\"odinger Evolution & Clock
Dephasing")
    axs[1].set_ylim(-0.58, 0.58)
```

```
    axs[1].legend(loc="upper right", fontsize=8)
    axs[1].grid(True, alpha=0.3)

    plt.tight_layout()
    plt.savefig("quadrature_time_simulation.pdf", dpi=300)
    plt.show()
\end{lstlisting}

%==========================================================================
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% REFERENCES
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